Aileron Trim Tab System – General Data

1. General

An electrically-operated aileron trim tab, hinged to the trailing edge of the outboard end of the left aileron, is controlled by an aileron trim switch mounted on the trim console.

2. Description

The aileron trim tab system consists of an aileron trim tab, actuator, switch, and position transmitter and indicator. Refer to Figure 1 for details.

A. Trim Tab

The aileron trim tab is hinged to the left aileron top skin at the trailing edge. A hinge arm, riveted to the lower surface of the tab, connects to a push-pull rod which attaches to a trim tab actuator.

B. Trim Tab Actuator

The aileron trim tab actuator is an electro-mechanical unit powered by a DC motor, which, when operating, derives its power from the DC bus. Limit switches in the actuator operate at both extremities of travel to switch off the motor.

C. Trim Tab Switch

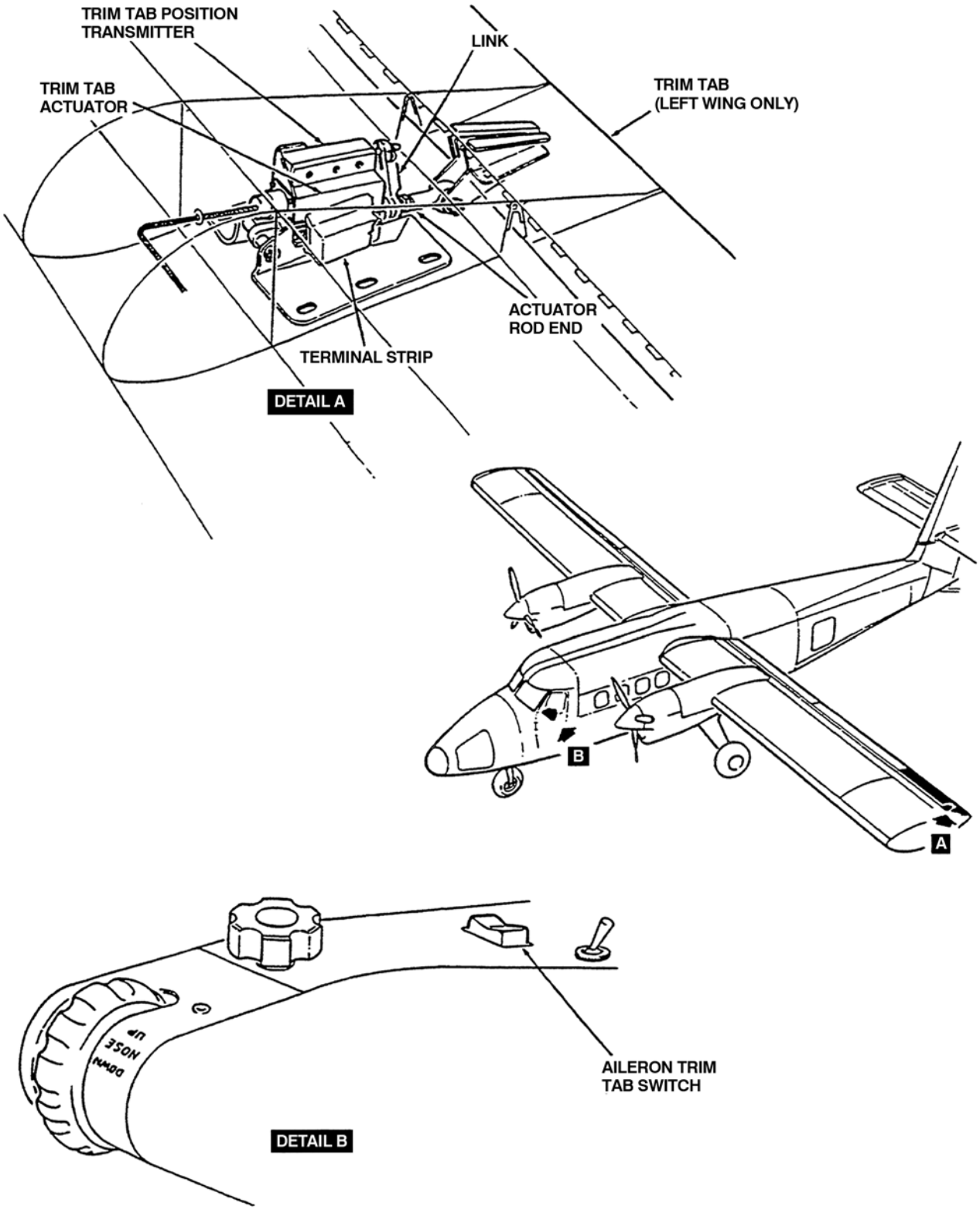
The aileron trim tab switch, mounted on the trim console, is marked LW DN and RW DN. It is a tilt type switch which, when released after making a trim selection, returns to its center off position to stop actuator movement at the selected position.

D. Trim Tab Position Transmitter

The aileron trim tab position transmitter installed on the trim tab actuator body, is an electro-mechanical unit which transmits aileron trim tab position to Apex, through the DAUs. The transmitter is a linear potentiometer with the contact arm connected to the actuator push-pull rod by a link. Extension or retraction of the actuator push-pull rod, causes a parallel movement of the contact arm resulting in a change of voltage being transmitted to the indicator.

E. Trim Tab Position Indicator

The aileron trim tab position is integrated into the APEX flight controls window and provides a visual indication of the trim tab position.



Aileron Trim Tab System
Figure 1

3. Operation

When the aileron trim tab switch is pressed to LW DN the actuator motor rotates to retract the actuator rod, which through the push-pull control rod, moves the trim tab down. The actuator will continue to retract until the actuator limit switch operates, or the trim switch is released; in either case the actuator will remain in the selected position until a further selection is made. During actuator rod movement, the position transmitter will have relayed the actuator position to Apex in respect of the amount of trim tab movement. The same sequence of events will occur with a RW DN selection, except that the actuator rod will extend, and the trim tab will move up.

4. Adjustment/Test

A. Rig Aileron Trim Tab System

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Wait several minutes for the aircraft electrical and avionics systems to initialize.

3) Operate aileron to neutral position.

4) Operate trim tab until trailing edges of aileron and trim tab align.

5) Check that Roll trim display indication on the Apex flight controls window reads centered.

6) If Roll trim display indication does not read centered, disconnect actuator rod from trim tab, operate actuator until centered, adjust actuator rod length and reconnect to trim tab.

Note If a zero reading is not obtained with the actuator in its mid-travel position, the trim tab position transmitter should be readjusted (refer to 27-13-21, Aileron Trim Tab Position Transmitter – Maintenance Practices, Para 2.).

7) Do a function test for aileron trim tab position potentiometer as per 27-13-21, Aileron Trim Tab Position Transmitter – Maintenance Practices, Para 3.A.)

8) Operate trim tab through its ranges of travel and check for full and free range of movement.

9) Check trim tab control surface clearances (refer to 27-00-00, Flight Controls – General Data).

10) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

5. Inspection/Check

A. Free Play Check – Aileron Trim Tab

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Operate ailerons to neutral position.

3) Operate trim tab to neutral position.

4) Move trim tab manually and check that free play (backlash), measured at trailing edge, does not exceed 0.10–inch.

Note Measurement to be taken adjacent to control rod attachment point.

5) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

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